

VWAP Reclaim Value Reclaim v2

Research Note and Strategy Whitepaper

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Prepared for TradingView Pine Script implementation and backtesting. This material is educational and is not financial advice.

Executive summary

This whitepaper documents the research logic behind the second version of the VWAP Reclaim Value Reclaim strategy. The original algorithm used a simple intraday value reclaim: price dips below session VWAP, then closes back above it. The v2 implementation keeps that hypothesis but adds context filters, a retest-confirmation state machine, volatility-normalized risk controls, and more conservative backtest assumptions.

The central design principle is that VWAP is a value and liquidity anchor, not a standalone predictor. A long setup should therefore require evidence of reclaimed value, renewed participation, market structure support, favorable regime, and acceptable initial R risk.

Research synthesis

Finding	Implication for the algorithm	Implemented in v2
VWAP is widely used as an execution benchmark and average traded price reference, but it is not inherently predictive by itself [R5, R6, R8].	Use VWAP as the value anchor, then require confirming evidence rather than buying every cross.	Minimum dip, bars below VWAP, reclaim buffer, candle quality, previous-high break, and no-chase rule.
Intraday volume is dynamic, and VWAP-related execution risk is linked to volume uncertainty and volume patterns [R6, R7].	A reclaim should be more credible when current participation is above normal.	Relative volume filter and optional lower-band sweep.
ADX measures trend strength and can help assess whether momentum is present [R10].	Filter out flat, low-conviction reclaims and align entries with positive directional movement.	ADX/DMI filter with optional +DI > -DI requirement.
Opening-range strategies and intraday breakout logic can be regime-sensitive; results may not be robust across subperiods [R11].	Use opening range as context rather than as a single alpha engine.	Optional OR midpoint or OR high filter.
Backtests are vulnerable to overfitting, selection bias, and unrealistic order-fill assumptions [R12, R13].	Make execution assumptions more conservative and provide a validation workflow.	Commission, slippage, Bar Magnifier, limit-fill verification, and a tutorial validation protocol.

Research implications by theme

Theme	Research takeaway	Strategy consequence
VWAP as benchmark	Execution literature treats VWAP as a benchmark tied to market volume and execution quality, not as a magic support/resistance line.	The script uses VWAP to define value, then requires acceptance and regime confirmation.
Volume participation	Intraday volume patterns change across the day and across symbols.	The algorithm adds RVOL and optional lower-band sweep so the reclaim is not purely price-based.
Trend strength	Trend-strength tools such as ADX can help distinguish active trends from flat markets.	ADX/DMI and VWAP slope are used to suppress weak reclaim attempts.
Opening range	Opening range logic may work in some regimes but can be unstable across subperiods.	OR is used only as structural confluence, not as the sole reason to enter.
Backtest reliability	Multiple trials, selection bias, and non-normal returns can inflate apparent performance.	The recommended workflow emphasizes out-of-sample testing, parameter stability, and cost sensitivity.

Base hypothesis

The strategy targets an intraday-to-swing long pattern. The market first trades below the session's volume-weighted value, creating a potential failed breakdown or liquidity sweep. A valid reclaim occurs when price closes back above VWAP with enough strength to suggest that sellers failed to maintain acceptance below value. The swing component is activated only when higher-timeframe and market-regime filters support continuation.

Formula

Session VWAP = $\text{sum}(\text{price}_i \times \text{volume}_i) / \text{sum}(\text{volume}_i)$. The implementation uses the selected source, default hlc3, and resets at the configured VWAP/value session. A weighted intraday standard deviation creates upper and lower VWAP bands.

Algorithm architecture

Stage	Rule stack
1. Value construction	Reset cumulative price-volume, volume, and price-squared-volume at the start of the selected session. Compute VWAP and weighted standard deviation bands.
2. Sweep tracking	Track the latest bar that dipped below VWAP by minDipPct and, optionally, the latest touch of the lower VWAP band.
3. Reclaim trigger	Require close crossing above VWAP, close above a VWAP buffer, enough bars below VWAP, no chase beyond maxChasePct, bullish candle quality, and optional previous-high break.
4. Structural context	Require opening range midpoint or high confirmation if enabled.
5. Regime context	Require HTF EMA trend, optional market proxy EMA, VWAP slope, ADX/DMI, RVOL, ATR%, and optional gap filter.
6. Entry mode	Immediate mode enters at the reclaim close. Retest-confirmed mode arms the setup, waits for a VWAP hold, and enters only after a break above the stored trigger high.
7. Risk gate	Compute stop anchor plus ATR buffer. Block trades whose initial risk percentage is too small or too large.
8. Exit engine	Scale out at TP1, move to breakeven, trail the runner by ATR, VWAP/ATR hybrid, or structure. Optional exits include HTF EMA loss, VWAP failure, max hold, and end-of-day flat.

State machine pseudocode

The production script is written in Pine v6, but the logic can be summarized as the following state machine.

```
For each intraday bar:
- Update session VWAP and VWAP deviation bands.
- Track opening range high, low, and midpoint.
- If price dips below VWAP by the configured threshold, store the dip bar.
- If price crosses back above VWAP and all reclaim, structure, regime, and risk filters pass, create a base reclaim event.
- If entry mode is Immediate, submit a long entry.
- If entry mode is Retest-confirmed, arm the setup, wait for VWAP hold, then enter on a close above the stored reclaim high.
- Once in trade, maintain initial stop, TP1, breakeven, active trailing stop, and optional signal exits.
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Non-repainting implementation details

Risk area	Implementation choice
Higher-timeframe trend	HTF close and EMA use confirmed historical values with expression offsets and barmerge.lookahead_on, matching TradingView guidance for non-repainting HTF requests.
Session boundaries	time(timeframe.period, sessionString) determines whether the current bar belongs to VWAP, entry, OR, or flat windows.

Risk area	Implementation choice
Close-confirmed signals	Entries are based on bar closes, not intrabar unconfirmed crosses. process_orders_on_close is enabled for close-based simulation.
Retest state	The script stores retest variables using var state, then clears them on new session, entry, timeout, or failed close below VWAP.
Risk state	The active stop only tightens for long positions. It does not intentionally loosen after entry.

Why each major change was made

- 1. Minimum time below VWAP.** A one-bar dip and immediate cross can be random noise. Requiring a minimum number of bars below VWAP forces the setup to represent a real failed acceptance below value.
- 2. Candle quality and previous-high break.** A close in the upper part of the bar and above the previous high indicates stronger buying pressure than a close barely above VWAP.
- 3. No-chase rule.** Because VWAP is a lagging intraday average, entering too far above VWAP can convert a reclaim idea into a momentum chase. The maxChasePct rule keeps the entry near the value anchor.
- 4. Opening range context.** The opening range often defines early intraday structure. Requiring a reclaim above the OR midpoint or high helps distinguish a constructive reclaim from one still trapped below early supply. Because ORB research can be regime-sensitive, this is used as a filter, not as the main signal.
- 5. Regime filters.** The strategy is long-only. Higher-timeframe EMA, market proxy EMA, VWAP slope, ADX/DMI, RVOL, and ATR% filters work together to avoid long setups when broader context is weak, participation is low, or volatility is unsuitable.
- 6. Retest-confirmed mode.** The immediate reclaim signal can work in sharp reversals, but it can also overtrade chop. Retest-confirmed mode sacrifices some speed to require proof that VWAP has flipped from resistance back into support.
- 7. R-based exit design.** Scaling out at TP1 and moving the stop toward breakeven converts the setup into a two-part trade: one part monetizes the reclaim impulse, while the runner attempts to capture a swing continuation.

Design alternatives considered

Alternative	Reason it was not the default
Buy every close above VWAP	Too many entries in chop; ignores whether price actually swept value and reclaimed it.
Require only lower VWAP band touch	This can be too strict for strong trend days. It remains optional for conservative profiles.
Enter only above opening range high	Cleaner but can miss midday value reclaims that work without clearing the full opening range.
Use RSI oversold as a primary trigger	RSI can help, but the strategy's thesis is value reclaim, not oscillator mean reversion. Adding RSI can be tested later as an optional module.
Always exit at fixed R target	Fixed targets can truncate swing runners. Version 2 supports fixed targets but defaults to a runner trail.
Use lower-timeframe request.security for intrabar signals	Lower-timeframe aggregation can complicate history and fill assumptions. This version stays close-confirmed on the chart timeframe.

Execution and backtest design

TradingView strategies are historical simulations. Version 2 uses process_orders_on_close so close-confirmed signals can be processed on the signal bar, Bar Magnifier for more granular historical fills when available, 1 tick of slippage, 0.01% commission, and limit-fill verification. These settings do not remove all simulation risk, but they reduce the tendency to overstate performance.

Backtest risk	Mitigation
Lookahead bias from higher timeframes	HTF values use confirmed past-bar expressions such as <code>close[1]</code> with <code>barmerge.lookahead_on</code> , following TradingView's non-repainting guidance [R2].
Unrealistic fill assumptions	Strategy declaration includes slippage, commission, Bar Magnifier, and limit verification [R4].
Over-optimization	Use wide parameter bands, out-of-sample periods, and sensitivity tests. Do not tune to the final holdout [R12, R13].
Regime dependence	Evaluate separately in trend, range, high-volatility, and low-volatility regimes.
Symbol selection bias	Test on a predefined universe rather than only on examples where the strategy looks good.

Metrics and formulas to report

TradingView provides many built-in metrics. For this strategy, R-based diagnostics are especially important because the initial stop distance changes from trade to trade.

Metric	Formula or definition	Reason to report
Initial R	Entry price - initial stop for long trades	Normalizes trade risk.
Trade R	Net trade profit / initial R risk	Compares trades with different stop distances.
Average R	Mean of trade R values	Shows expectancy independent of share size.
Median R	Median of trade R values	Reduces distortion from one large runner.
Max adverse excursion	Worst open loss during trade in R	Shows whether stops are too tight or too loose.
Exposure time	Bars or sessions in position	Important for intraday-to-swing holding risk.
Cost sensitivity	Net performance after increasing slippage/commission	Tests whether edge depends on perfect fills.

Validation framework

- 1 Define the universe before looking at results. Example: liquid large-cap equities, index ETFs, or a specific futures market.
- 2 Choose a timeframe family. Example: 5-minute for entries and daily for trend filter. Avoid mixing results from many timeframes without reporting the number of trials.
- 3 Split the sample into development, walk-forward, and final holdout windows. Do not alter rules after reviewing final holdout performance.
- 4 Report R-based metrics in addition to TradingView net profit: average R, median R, largest loss in R, profit factor, drawdown in R, exposure time, and consecutive losses.
- 5 Run parameter stability checks. Robustness is indicated when nearby settings remain acceptable rather than one exact combination producing all performance.
- 6 Conduct cost sensitivity. Double slippage and commission, then check whether the edge disappears.
- 7 Forward test with alerts. Compare actual fills to simulated fills before any live deployment.

Recommended reporting template

Report field	Example values to include
Universe	Symbols tested, liquidity filters, and whether proxy filter was enabled.
Timeframe	Chart timeframe, VWAP session, entry session, and HTF trend timeframe.
Date windows	Development, walk-forward, and final holdout windows.
Costs	Commission, slippage, Bar Magnifier status, and limit-fill verification.
Parameters	Only final parameter ranges or chosen values, not every failed experiment.
Performance	Net profit, max drawdown, profit factor, average R, median R, trade count, and exposure.
Robustness	Sensitivity table for minDipPct, minRvol, minAdx, maxChasePct, and entryMode.
Failure modes	Largest losses, worst regimes, and examples of false reclaims.

Future research roadmap

- Add a time-of-day RVOL model that compares 10:00 volume to historical 10:00 volume instead of using a simple volume SMA.
- Add optional anchored VWAPs from prior day high, prior day low, earnings gaps, or session swing lows.
- Add a short-side mirror strategy only after long-side validation is complete; short VWAP reclaims often have different behavior and borrow/liquidity issues in equities.
- Export closed trades to Python or R to compute R distributions, bootstrapped confidence intervals, probability of backtest overfitting proxies, and walk-forward tables.
- Build symbol-universe tests to reduce cherry-picking and identify where the pattern is most structurally suitable.

Parameter sensitivity map

Parameter	Lower value tends to	Higher value tends to
minDipPct	Increase signals but admit shallow noise	Require deeper value sweep and reduce trade count
minBarsBelowVwap	Capture faster reversals	Reduce whipsaw reclaims
maxChasePct	Avoid extended entries but miss fast momentum	Capture stronger moves but risk buying late
minRvol	Allow quieter names	Require stronger participation
minAdx	Allow range recoveries	Favor trend-strength conditions
openingRangeMode	More trades when Off	More structure when Above OR Mid or Above OR High
entryMode	Immediate: more trades	Retest: fewer, more confirmed trades
stopAtrMult	Tighter stops, higher stopout risk	Wider stops, lower size in risk-based mode

Limitations

- VWAP is an intraday value reference and lagging measure. It should not be treated as a predictive model by itself.
- The script uses OHLCV chart data, not order-book liquidity, hidden orders, news, or true tick-by-tick execution data.
- TradingView backtests are simulations. Bar Magnifier improves historical fill modeling only when lower-timeframe data is available.

- The market proxy filter is useful for many US equity workflows, but it can be inappropriate for futures, crypto, FX, or non-US symbols.
- Risk-based sizing uses strategy equity, stop distance, and syminfo.pointvalue. Instrument-specific lot size constraints may still need manual adjustment.
- No historical performance claim is made in this document. The whitepaper provides an implementation and validation framework, not a guarantee of profitability.

Implementation summary

File	Purpose
vwap_reclaim_value_reclaim_v2.pine	Full Pine Script v6 strategy implementation with all v2 upgrades.
VWAP_Reclaim_Value_Reclaim_v2_Tutorial.pdf	Operating manual, installation steps, input guide, presets, testing workflow, and troubleshooting.
VWAP_Reclaim_Value_Reclaim_v2_Whitepaper.pdf	Research note, architecture, design rationale, validation protocol, and references.

References

ID	Source	Use in this work
R1	TradingView Pine Script v6 User Manual - Strategies https://www.tradingview.com/pine-script-docs/concepts/strategies/	Pine strategies simulate trades across historical and realtime bars and support backtesting, order placement, and performance analysis.
R2	TradingView Pine Script v6 User Manual - Other timeframes and data https://www.tradingview.com/pine-script-docs/concepts/other-timeframes-and-data/	TradingView documents that non-repainting higher-timeframe requests should use confirmed past-bar expressions, such as <code>close[1]</code> , with <code>barmerge.lookahead_on</code> .
R3	TradingView Pine Script v6 User Manual - Sessions https://www.tradingview.com/pine-script-docs/concepts/sessions/	<code>time()</code> can test whether bars are inside time-based sessions; <code>input.session()</code> lets users configure session times.
R4	TradingView Pine Script v6 User Manual - Strategy order fills, Bar Magnifier, slippage https://www.tradingview.com/pine-script-docs/concepts/strategies/	TradingView documents Bar Magnifier, <code>process_orders_on_close</code> , slippage, and order-fill assumptions for more realistic strategy simulations.
R5	Konishi, H. - Optimal slice of a VWAP trade https://www.sciencedirect.com/science/article/abs/pii/S1386418101000234	VWAP is described as volume-weighted average price over a trading period and as an industry-standard benchmark.
R6	Busseti, E. and Boyd, S. - Volume Weighted Average Price Optimal Execution https://arxiv.org/abs/1509.08503	VWAP is studied as an execution benchmark, and volume uncertainty is a central execution problem.
R7	Bialkowski, J., Darolles, S., and Le Fol, G. - Improving VWAP strategies: A dynamic volume approach https://www.sciencedirect.com/science/article/abs/pii/S0378426607003226	The paper argues that modeling intraday volume dynamics can reduce VWAP execution risk.
R8	Investopedia - Volume-Weighted Average Price https://www.investopedia.com/terms/v/vwap.asp	VWAP is a single-day, lagging indicator; it is useful for liquidity and average traded price context but is not predictive by itself.
R9	IG - Relative Volume Indicator https://www.ig.com/uk/trading-strategies/what-is-the-relative-volume-indicator-and-how-do-you-use-it-when-230904	Relative volume compares current stock volume with average traded volume; high RVOL can signal more liquidity and volatility potential.
R10	Fidelity - Average directional index: ADX https://www.fidelity.com/viewpoints/active-investor/average-directional-index-ADX	ADX is used to evaluate trend strength, which can help determine whether momentum is behind a move.
R11	Holmberg, U., Lonnroth, J., and Lundstrom, C. - Assessing the profitability of intraday opening range breakout strategies https://www.sciencedirect.com/science/article/abs/pii/S1544612312000438	The paper reports that opening-range breakout performance was not robust across subperiods, supporting careful regime testing.
R12	Bailey, D. H., Borwein, J., Lopez de Prado, M., and Zhu, Q. J. - The Probability of Backtest Overfitting https://papers.ssrn.com/sol3/papers.cfm?abstract_id=2326253	The authors present a framework for estimating the probability that investment backtests are overfit.
R13	Bailey, D. H. and Lopez de Prado, M. - The Deflated Sharpe Ratio https://papers.ssrn.com/sol3/papers.cfm?abstract_id=2460551	The authors describe selection bias and non-normal returns as sources of inflated performance expectations.